WExperiment 6: BJT Amp Biasing

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Objective: To study the operating points of a BJT amplifier by studying the methods of properly setting it. To learn the different types of biasing methods, including resistive divider biasing, emitter degeneration, simple biasing, and self-biasing.

Pre-Lab:

Experiment:

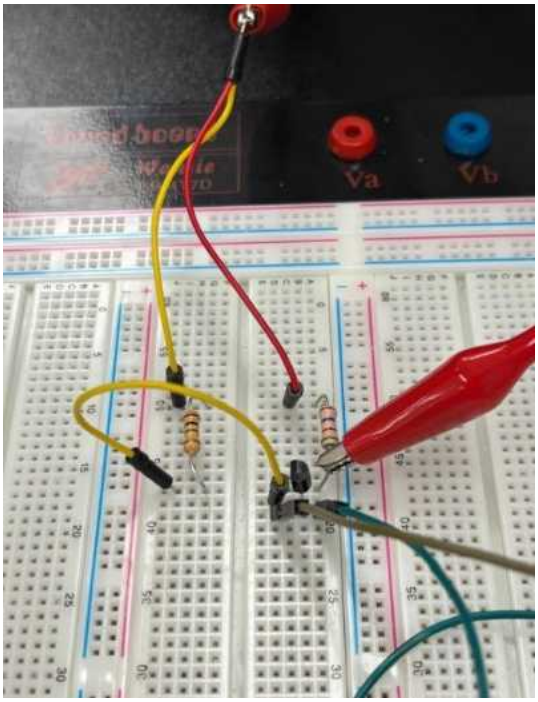
3.1 Simple biasing

Procedure:

1. Set up the circuit with R_B=1MΩ and R_C as 2.7kΩ.
2. Applied DC voltage to 12V and measure VBC, VCE, IB and IC.
3. Calculate beta value.
4. The RB is then changed and repeated for data collections.

	R_B=1MΩ	R_B=570kΩ
	Q1	Q1
V_{BE} [V]	0.61	0.61
V_{CE} [V]	9.28	9.06
I_B [μA]	11.41	25.4
I_C [mA]	0.94	1.02
β	82.38	40.15
Operating region		

Circuit:



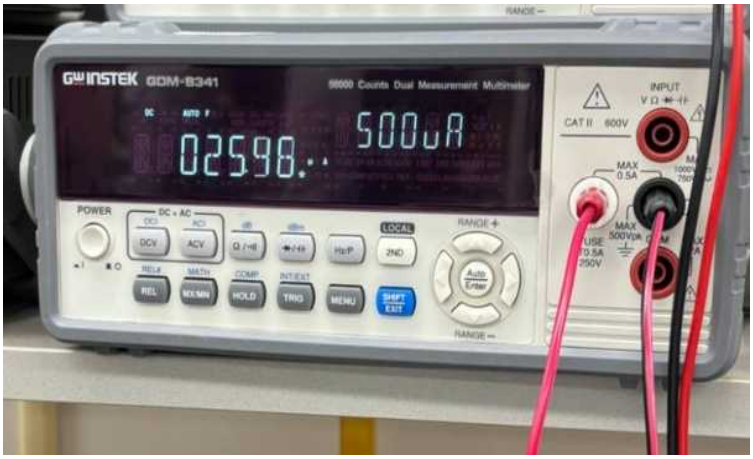
Vce when R_B=1MΩ.



I_B when R_B=1MΩ.



I_B when R_B=570kΩ.



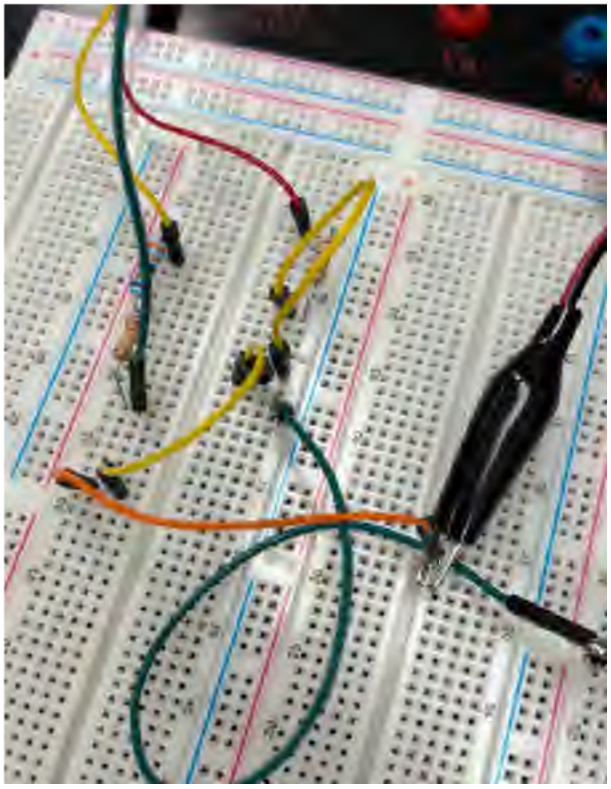
3.2 Resistive divider biasing

	R_2=8.2kΩ
V_{BE} [V]	0.61
V_{CE} [V]	9.14
I_B [μA]	95.7
I_C [mA]	0.984
β	
Operating region	

	R_2[kΩ]	
	6.8	12
$R_2/(R_1+R_2)$	0.053	0.090
I_C [mA]	95.5	95.5

V_{BE} [V]	0.61	0.61
V_{CE} [V]	9.4	8.80
Operating region		

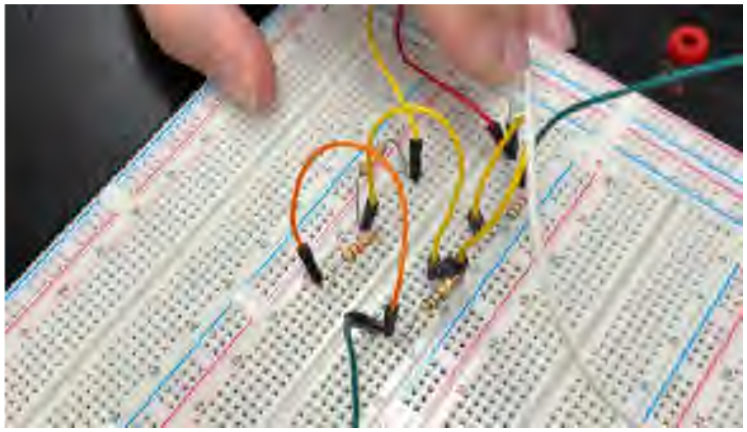
Circuit:



3.3 Resistive divider biasing with emitter degeneration

	R_2[kΩ]		
	6.8	8.2	12
$R_2/(R_1+R_2)$	0.053	0.063	0.090
I_C [mA]	0.347	0.526	0.724
V_{BE} [V]	0.766	1.30	1.02
V_{CE} [V]	0.992	1.48	2.048
Operating region			

Circuit:



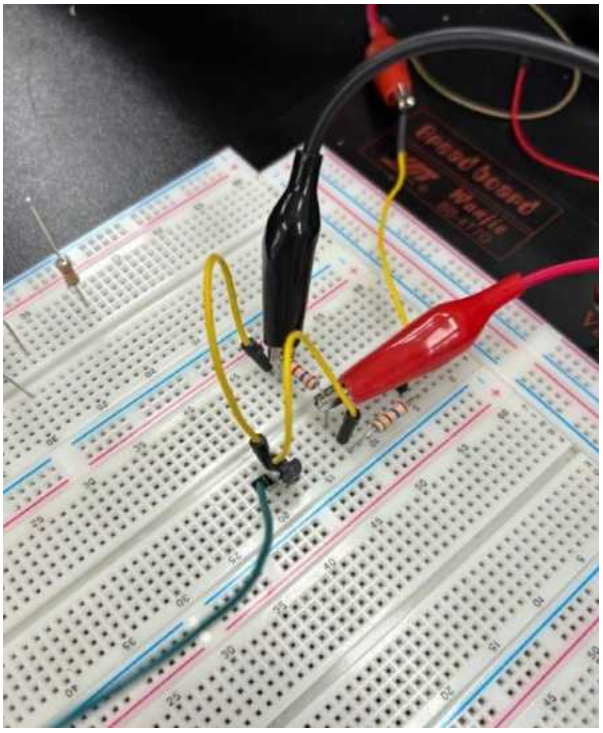
I_C when R_2=8.2kΩ.



3.4 Self biasing

	R_B =2.7kΩ	R_B=120kΩ	R_B=330kΩ
	Q1	Q1	Q1
V_{BE} [V]	1.55	7.87	8.44
V_{CE} [V]	9.73	3.46	2.90
I_B [μA]	558	66.9	25.7
I_C [mA]	3.43	1.22	1.02
β	6.14	18.23	39.68
R_B /β	439.73	6582.55	8316.53
Operating region			

Circuit:

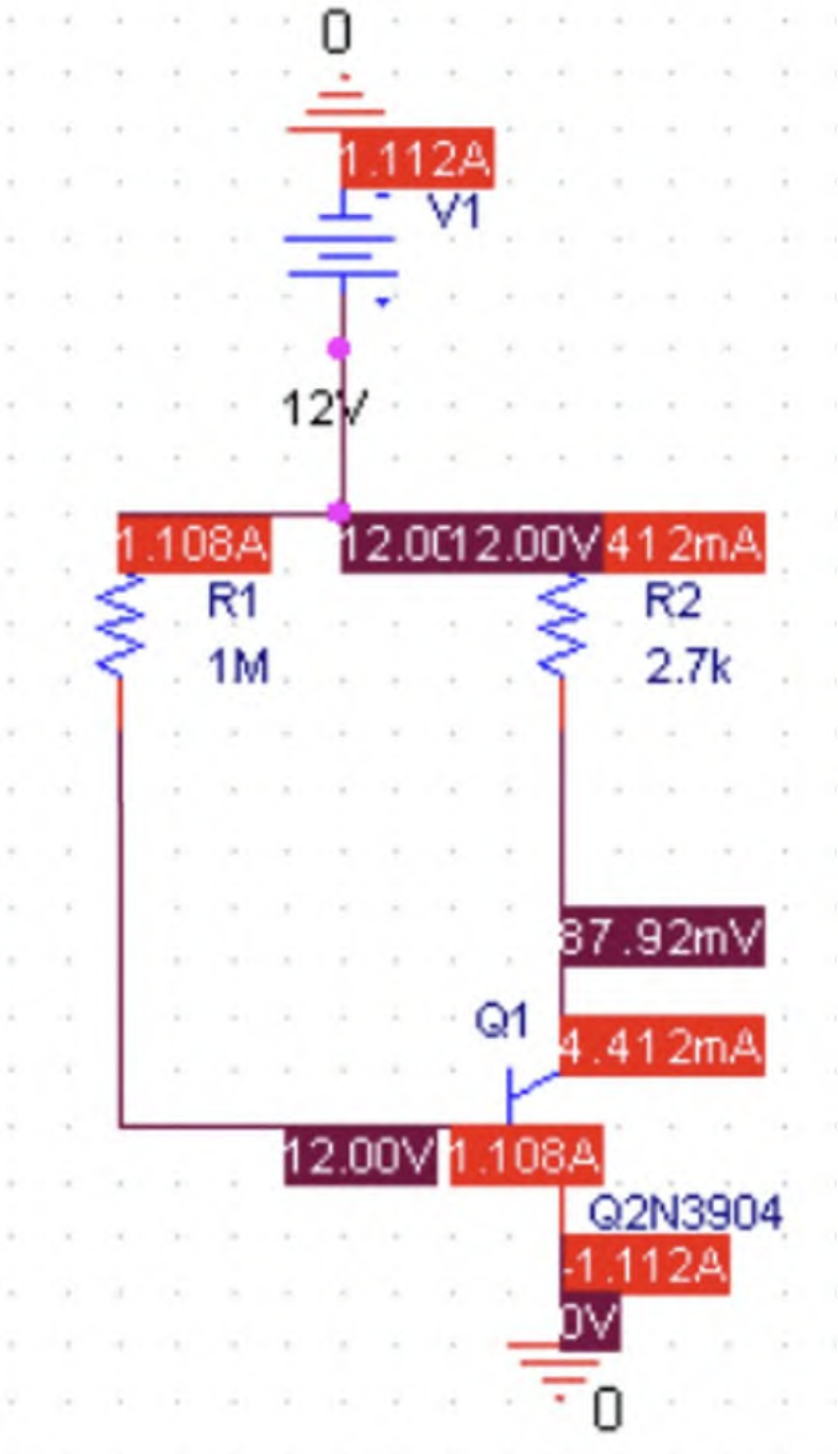


V_{BE} when R_B=2.7kΩ.

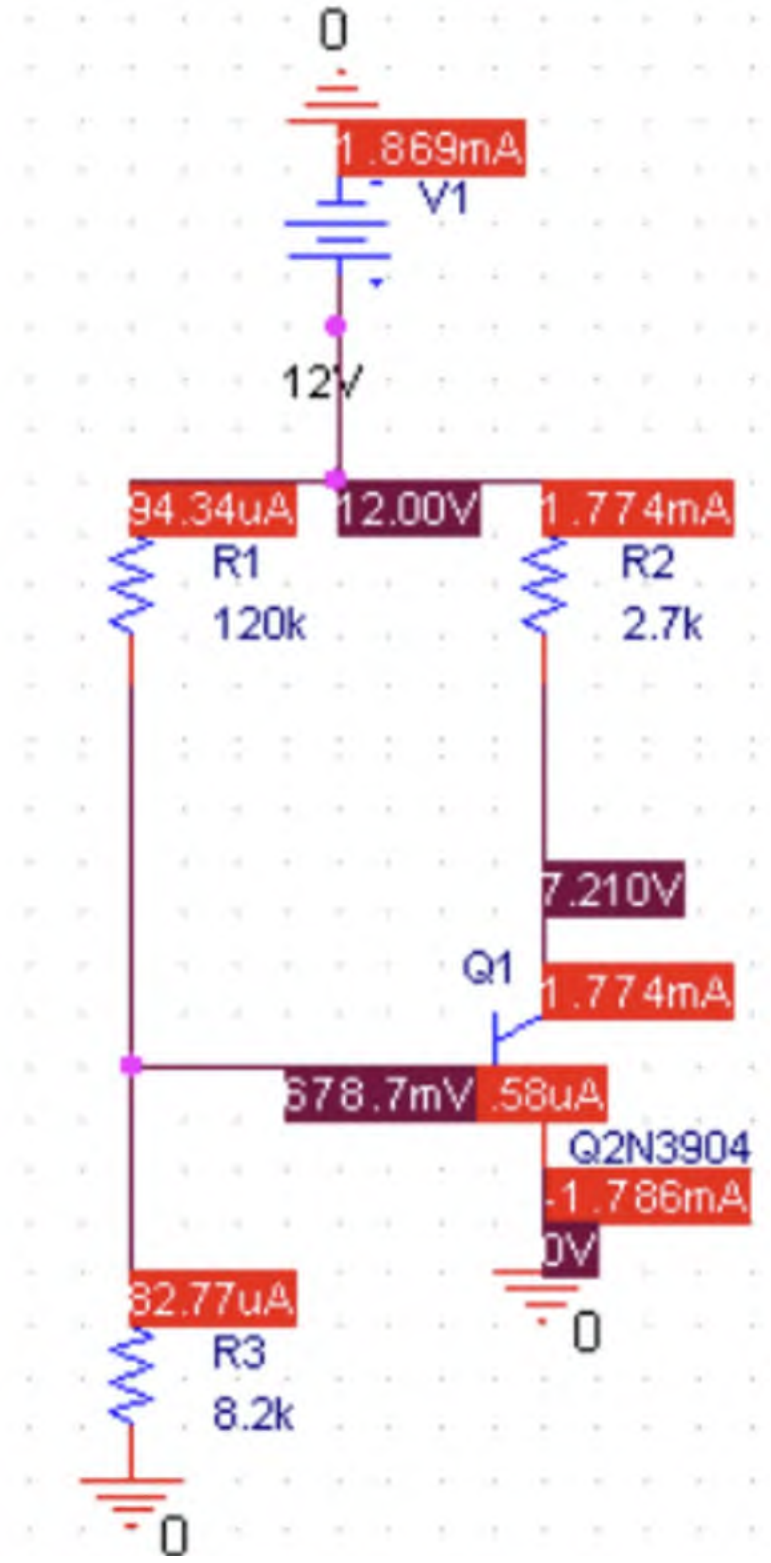


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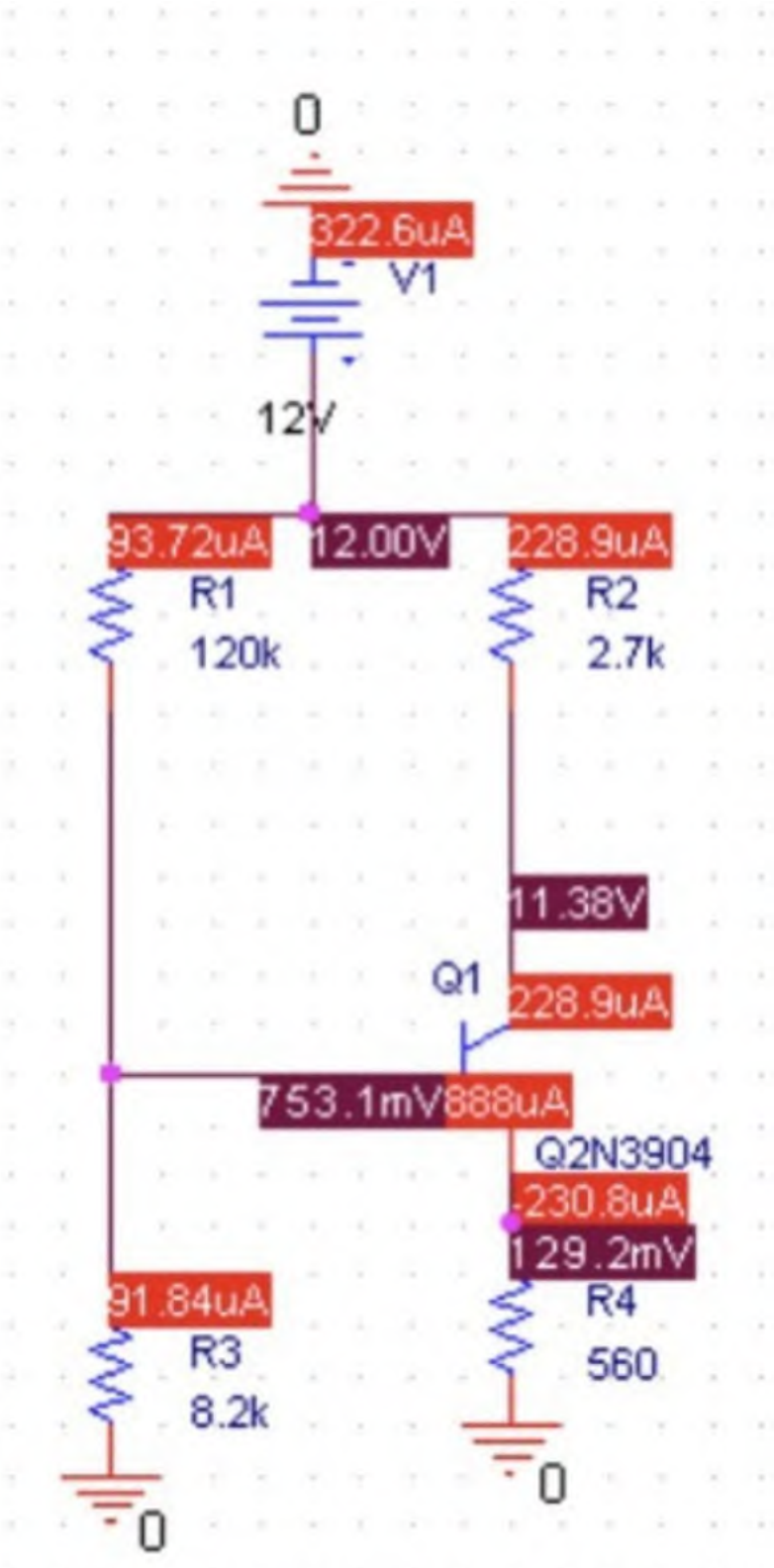
Exp. 3.1



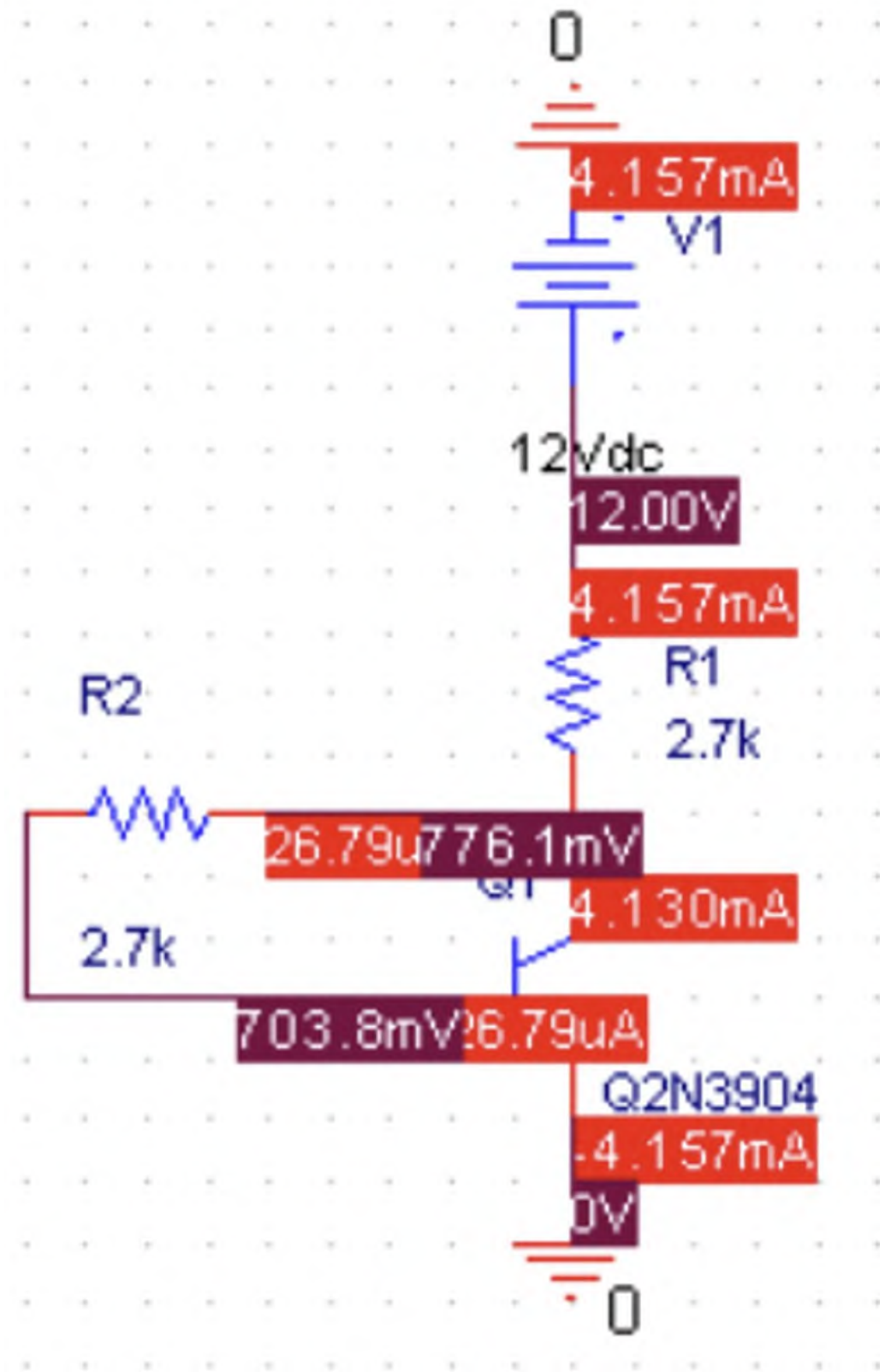
Exp. 3.2


0
1.869mA
V1
12V
94.34uA
12.00V
1.774mA
R1
120k
R2
2.7k
7.210V
Q1
1.774mA
678.7mV
.58uA
Q2N3904
-1.786mA
0V
82.77uA
R3
8.2k
0
0

Exp. 3.3


0
322.6uA
V1
12V
93.72uA
12.00V
228.9uA
R1
120k
R2
2.7k
11.38V
Q1
228.9uA
753.1mV
888uA
Q2N3904
230.8uA
129.2mV
91.84uA
R3
8.2k
R4
560
0
0

Exp. 3.4


0
4.157mA
V1
12Vdc
12.00V
4.157mA
R1
2.7k
R2
26.79u
776.1mV
Q1
4.130mA
2.7k
703.8mV
26.79uA
Q2N3904
-4.157mA
0V
0